

Exp. No: 2 Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

To write program for electric field and electric flux density in sphere using SCILAB. **Software required:**

- SCILAB version 6.0.1

Code:

```
EXP 1 EME.sci  Xp 1).sci  exp 1)ime.sci  exp 2.sci  exp 2 case 1.sci
1  clc;
2  clear;
3
4  q=input('Enter the value of q=');
5  r=input('Enter the value of r=');
6
7  efcilon=8.854*10^-12;
8  pi=3.14;
9  E=q/(4*pi*efcilon*r^2);
10 disp('The electric field is', [E]);
11 D=E*efcilon;
12 disp('The flux density is', [D]); x = linspace(0, 10, 50);
13 y = E * exp(-0.1 * x);
14 z = D * exp(-0.1 * x);
15
16 figure();
17 subplot(2, 2, 1);
18 plot2d3(x,y);
19 xlabel('R');
20 ylabel('Electric field intensity');
21 title('Electric field intensity');
22 subplot(2, 2, 2);
23 plot2d3(x,z);
24 xlabel('R');
25 ylabel('Electric flux intensity');
26 title('Electric flux intensity');
27
```

Output:

```
Scilab 2026.1.0 Console
File Edit Control Applications ?

File Browser
C:\Users\user\Documents\

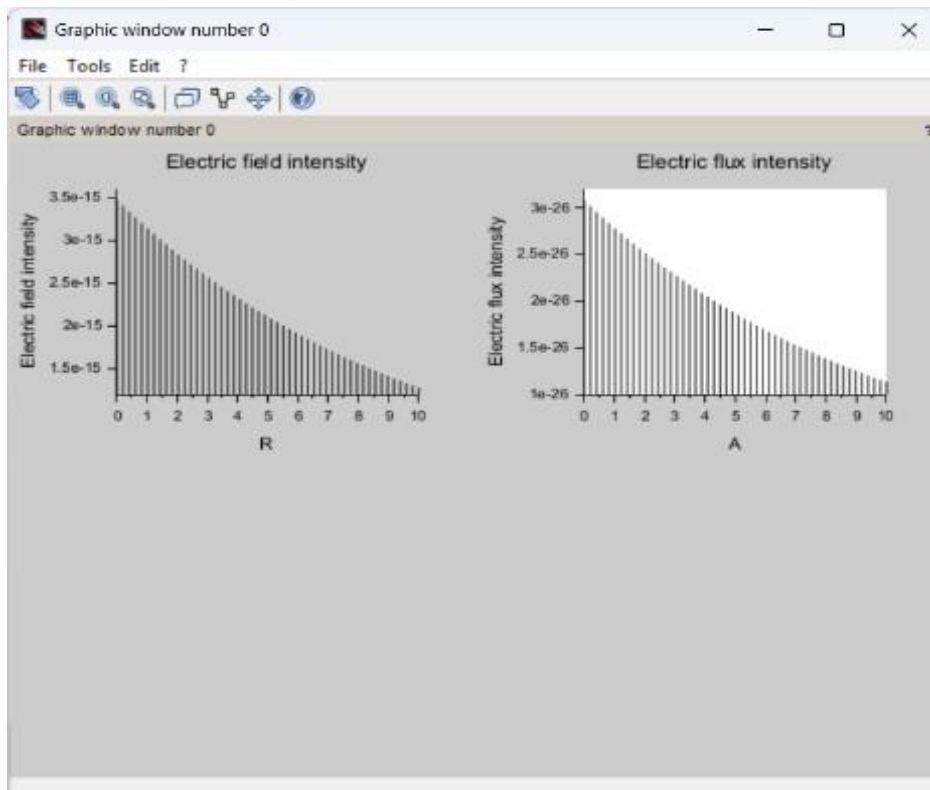
Name
Documents
..
My Music
My Pictures
My Videos
ECA05 EXP 2.sci~
ex 6.sc

--> exec('C:\Users\user\OneDrive\Documents\EX-2(I).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EX-2(I).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci', -1)
Enter the value of q = exec('C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci',
Warning: file 'C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci' already opened
Enter the value of q = 10

Enter the value of r = exec('C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci',
Warning: file 'C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci' already opened
Enter the value of q = 5*10^-6

Enter the value of r = 10

"The electric field intensity is:"
3.955D+09
"E = "
"The electric flux density is:"
0.0350141
"D = "
```



Theoretical calculation:

EXP NO. : 2

Determination and plotting of Electric field and Electric flux density in Sphere full of charges with varying radius.

Case (i)

Output :

Enter the value of $q = 4 \times 10^{-6}$

Enter the value of $r = 5 \text{ cm}$.

Electric flux Density (D) vs Radius graph is generated Successfully.

Case (ii)

Output :

Enter the value of $q = 5 \times 10^{-6}$

Enter the value of $r = 3 \text{ cm}$

Linear Density : D vs Radius graph is successfully generated.

Case (iii)

Output :

Enter the value of $q = 5 \times 10^{-6}$

Enter the value of $r = 10 \text{ cm}$

Electric flux Density (D) vs Radius graph is generated successfully.

Case (iv) :

Output :

Enter the value of $q = 3 \times 10^{-6}$

Enter the value of $r = 1 \text{ mm}$

Case 1:

Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0 .

What are the values of E and D at the center and at the surface of the sphere?

Code:

ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes

File Edit Format Options Window Execute ?

ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes

ECA05 EXP 2.sci

```
1 clear;
2 clear;
3
4 // Uniform charge density
5 rho0 = 1e-6; % Charge density (C/m^3)
6
7 // Permittivity of free space
8 epsilon0 = 8.854e-12; % F/m
9
10 // Radius of the sphere
11 R = 1; % m
12
13 // Number of points
14 n = 100;
15
16 // Radial distance
17 r = linspace(0, R, n);
18
19 // Initialise arrays
20 D = zeros(1, n);
21 E = zeros(1, n);
22
23 // Calculate D and E inside the uniformly charged sphere
24 for i = 1:n
25     ri = r(i);
26
27     if ri == 0 then
28         D(i) = 0;
29         E(i) = 0;
30     else
31         // Electric flux density
32         D(i) = (rho0 * ri) / 3;
33
34         // Electric field intensity
35         E(i) = D(i) / epsilon0;
36     end
37 end
38
39 end
```

ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes

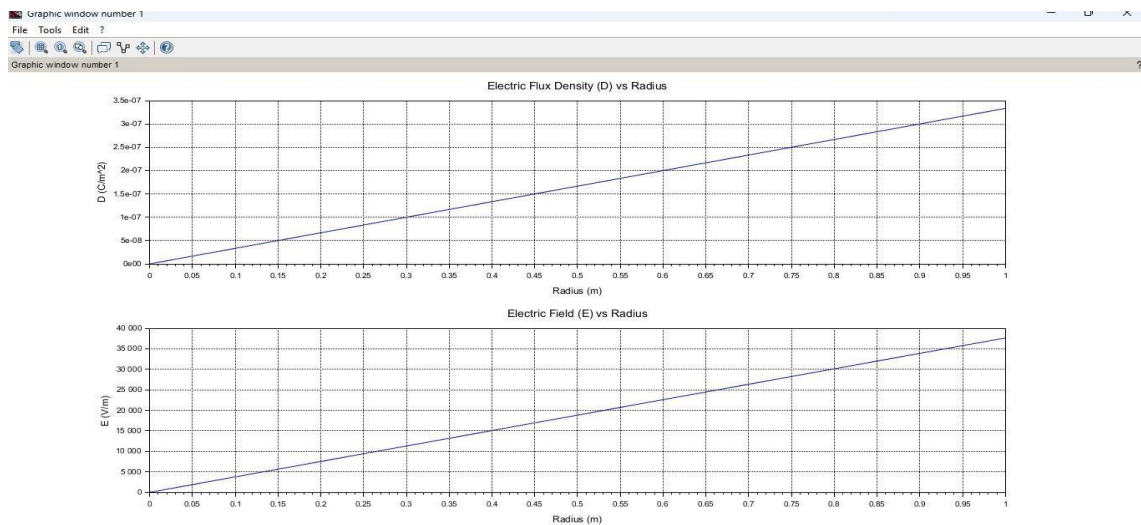
File Edit Format Options Window Execute ?

ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes

ECA05 EXP 2.sci

```
25 ri = r(i);
26
27 if ri == 0 then
28     D(i) = 0;
29     E(i) = 0;
30 else
31     // Electric flux density
32     D(i) = (rho0 * ri) / 3;
33
34     // Electric field intensity
35     E(i) = D(i) / epsilon0;
36 end
37
38 end
39
40 // Create figure
41 scf(0);
42 clf();
43
44 // Plot D vs Radius
45 subplot(1, 2, 1);
46 plot(r, D);
47 xlabel("Radius (m)");
48 ylabel("Electric Flux Density D (C/m^2)");
49 title("Uniform Charge Density: D vs Radius");
50 xgrid();
51
52 // Plot E vs Radius
53 subplot(1, 2, 2);
54 plot(r, E);
55 xlabel("Radius (m)");
56 ylabel("Electric Field E (V/m)");
57 title("Uniform Charge Density: E vs Radius");
58 xgrid();
59
60
```

Output:



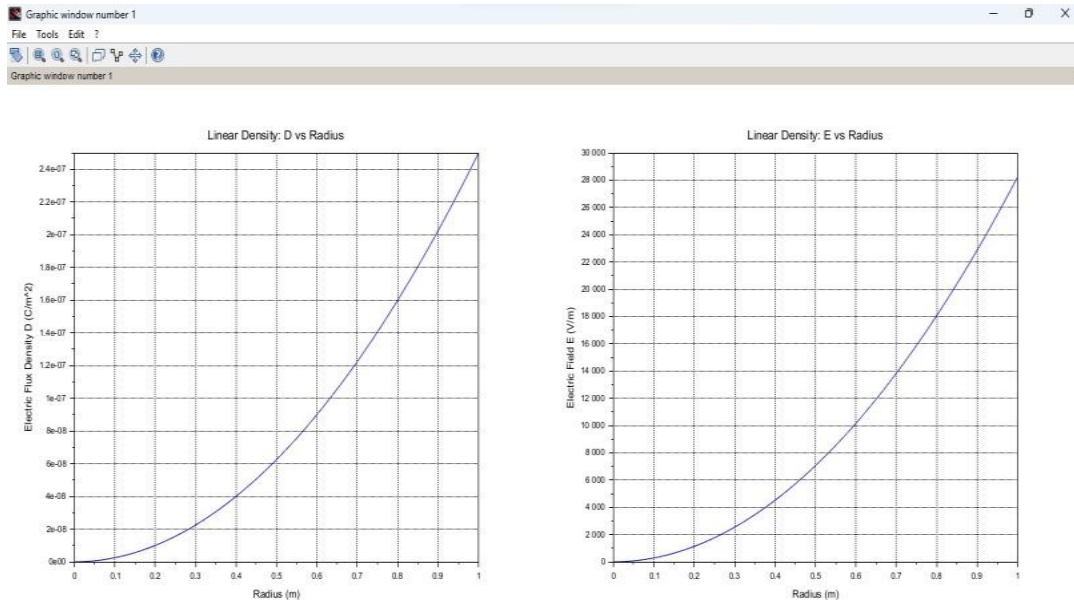
Case 2:

For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.

Code:

```
ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes
File Edit Format Options Window Execute ?
ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes
"ECA05 EXP 2.sci"
1 clc;
2 clear;
3
4 // Parameters
5 rho0 = 1e-6; // Coefficient of charge density (C/m^4)
6 epsilon0 = 8.854e-12; // Permittivity of free space (F/m)
7 R = 1; // Radius of sphere (m)
8 n = 100; // Number of points
9
10 // Radial distances
11 r = linspace(0, R, n);
12
13 // Initialize arrays
14 D = zeros(1, n);
15 E = zeros(1, n);
16
17 // Calculate D and E
18 for i = 1:n
19     ri = r(i);
20
21     if ri == 0 then
22         D(i) = 0;
23         E(i) = 0;
24     else
25         // Enclosed charge
26         Q_enclosed = rho0 * pi * ri^4;
27
28         // Electric flux density
29         D(i) = Q_enclosed / (4 * pi * ri^2);
30
31         // Electric field intensity
32         E(i) = D(i) / epsilon0;
33     end
34 end
35
36 // Create figure
37 figure;
38
39 // Plot D vs Radius
40 subplot(1, 2, 1);
41 plot(r, D);
42 xlabel("Radius (m)");
43 ylabel("Electric Flux Density D (C/m^2)");
44 title("Linearly-Increasing-Density: D-vs-Radius");
45 xgrid();
46
47 // Plot E vs Radius
48 subplot(1, 2, 2);
49 plot(r, E);
50 xlabel("Radius (m)");
51 ylabel("Electric Field E (V/m)");
52 title("Linearly-Increasing-Density: E-vs-Radius");
53 xgrid();
54
55 Backup finished...
```

output:



Case 3:

A sphere of radius $R=2$ m has a uniform charge density $\rho=3 \times 10^{-6}$ C/m³

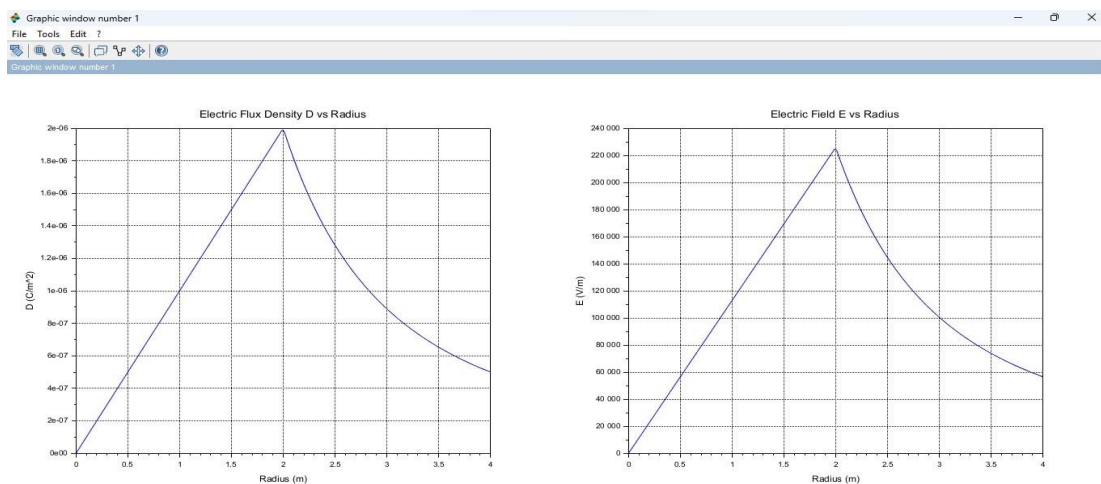
- 1. Calculate and plot the electric flux density $D(r)$ and electric field $E(r)$ as a function of radial distance r , ranging from 0 to 4 meters.**
- 2. Discuss the behaviour of the electric field inside and outside the sphere.**

Code:

```
ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes
File Edit Format Options Window Execute ?
[Icons]
ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes
ECA05 EXP 2.sci X

1 rho0 = 3e-6; // Uniform charge density (C/m^3)
2 epsilon = 8.854e-12; // Permittivity of free space (F/m)
3 R = 2; // Radius of the sphere (m)
4 n = 200; // Number of steps for plotting
5 r = linspace(0, 4, n); // Radial distances from 0 to 4 meters
6 // Initialise arrays for D and E
7 D = zeros(1, n);
8 E = zeros(1, n);
9 // Calculate the total charge enclosed in the sphere
10 Q_total = rho0 * (4 * pi * R^3 / 3);
11 // Calculate D and E
12 for i = 1:n
13     ri = r(i); // Current radius
14     // Enclosed charge for r <= R (inside the sphere)
15     if ri <= R then
16         Q_enclosed = rho0 * (4 * pi * ri^3 / 3);
17         // Enclosed charge for r > R (outside the sphere)
18     else
19         Q_enclosed = Q_total;
20     end
21     // Electric Flux Density D
22     D(i) = Q_enclosed / (4 * pi * ri^2);
23     // Electric Field E = D / epsilon
24     E(i) = D(i) / epsilon;
25 end
26 // Plotting
27 subplot(1, 2, 1);
28 plot(r, D, 'r');
29 label('r', 'Radius (m)');
30 label('D', 'Electric Flux Density (D) [C/m^2]');
31 title('Uniform Charge Density: D vs. Radius');
32 subplot(1, 2, 2);
33 plot(r, E, 'r');
34 label('r', 'Radius (m)');
35 label('E', 'Electric Field (E) [V/m]');
36 title('Uniform Charge Density: E vs. Radius');
37
```

output:



Result :

Thus the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.

